IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- SpaceX has gained worldwide attention for a series of historic milestones.

- It is the only private company ever to return a spacecraft from low-earth orbit, which it first accomplished in December 2010.
- SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars whereas other providers cost upward of 165 million dollars each, much of the savings is because Space X can reuse the first stage.
- Therefore, if we can determine if the first stage will land, we can determine the cost of a launch.
- This information can be used if an alternate company wants to bid against SpaceX for a rocket launch.
- This dataset includes a record for each payload carried during a SpaceX mission into outer space.

Introduction

- SpaceX would like to make predictions about the success/failure of SpaceX Falcon 9 rocket first stage landings.
- We want to know the nature and extent of the data that we have on SpaceX Falcon 9 first stage landings?
- What machine learning model could yield the highest accuracy in forecasting the success of a Falcon 9 first stage landing in an upcoming launch?

Section 1

Methodology

Methodology

Executive Summary

- • Collected data from SpaceX API and Wikipedia launch tables.
- • Prepared and refined data for visualizations, queries, and the development of machine learning models.
- • Conducted exploratory data analysis (EDA) through visualization techniques and SQL.
- • Generated interactive visual analytics utilizing Folium and Plotly Dash.
- • Performed predictive analysis using classification models

Data Collection

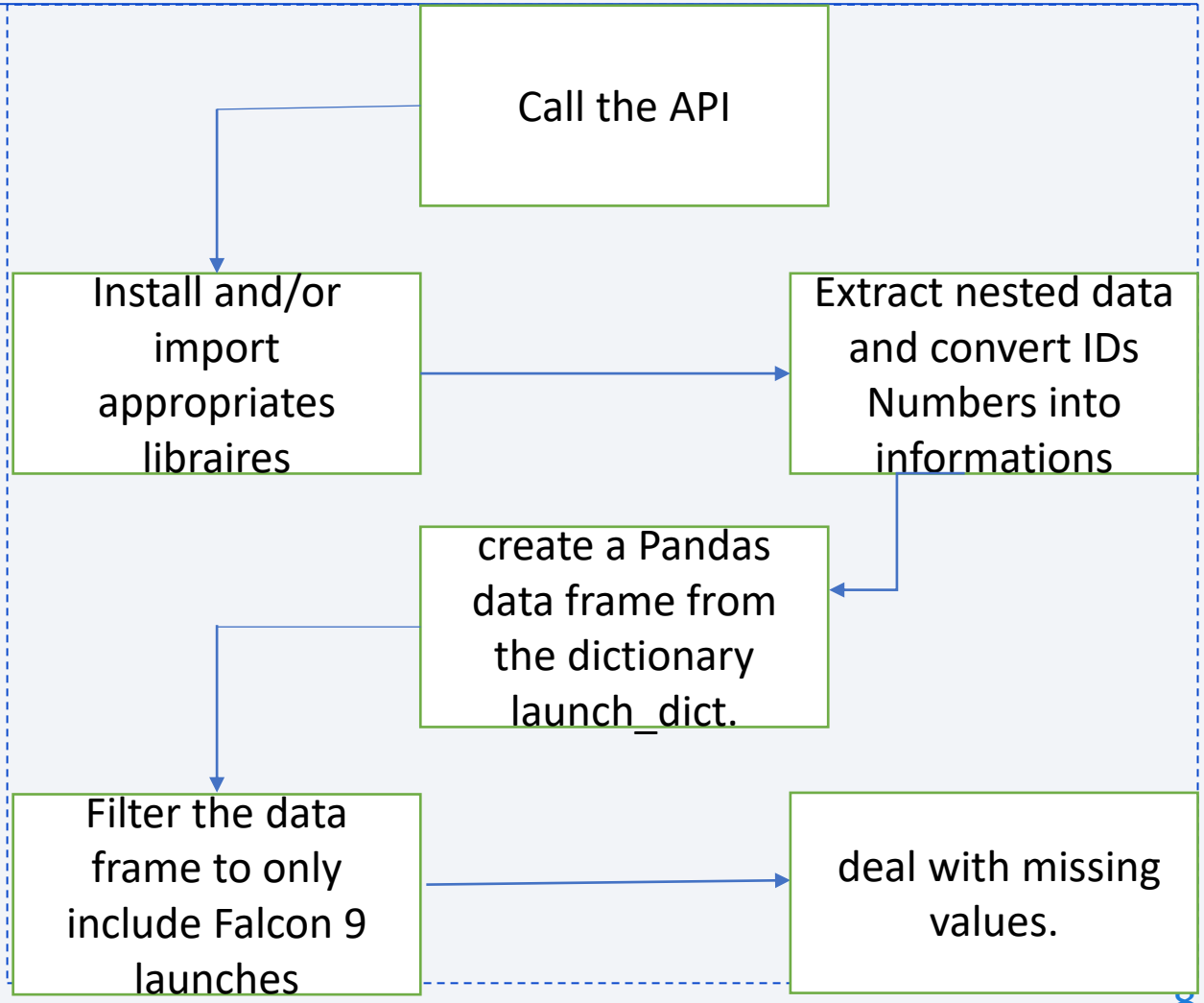
- The information and collect of data was assembled from diverse origins:
- one being an IBM mirror of the publicly available SpaceX API containing launch data in JSON format, another being an ongoing Wikipedia page with launch details in HTML tables (as of the revision made on June 9, 2021),
- and supplementary datasets were furnished, denoted by the darker shaded green .csv files positioned atop the diagram below.

Data Collection – SpaceX API

- Present your data collection with SpaceX REST has data publicly accessible. So that we request it and transform it in panda Data frame

- GitHub URL of Data Collection :

https://github.com/williamngoy/testrepo/blob/main/data-collection-api%20(1).ipynb

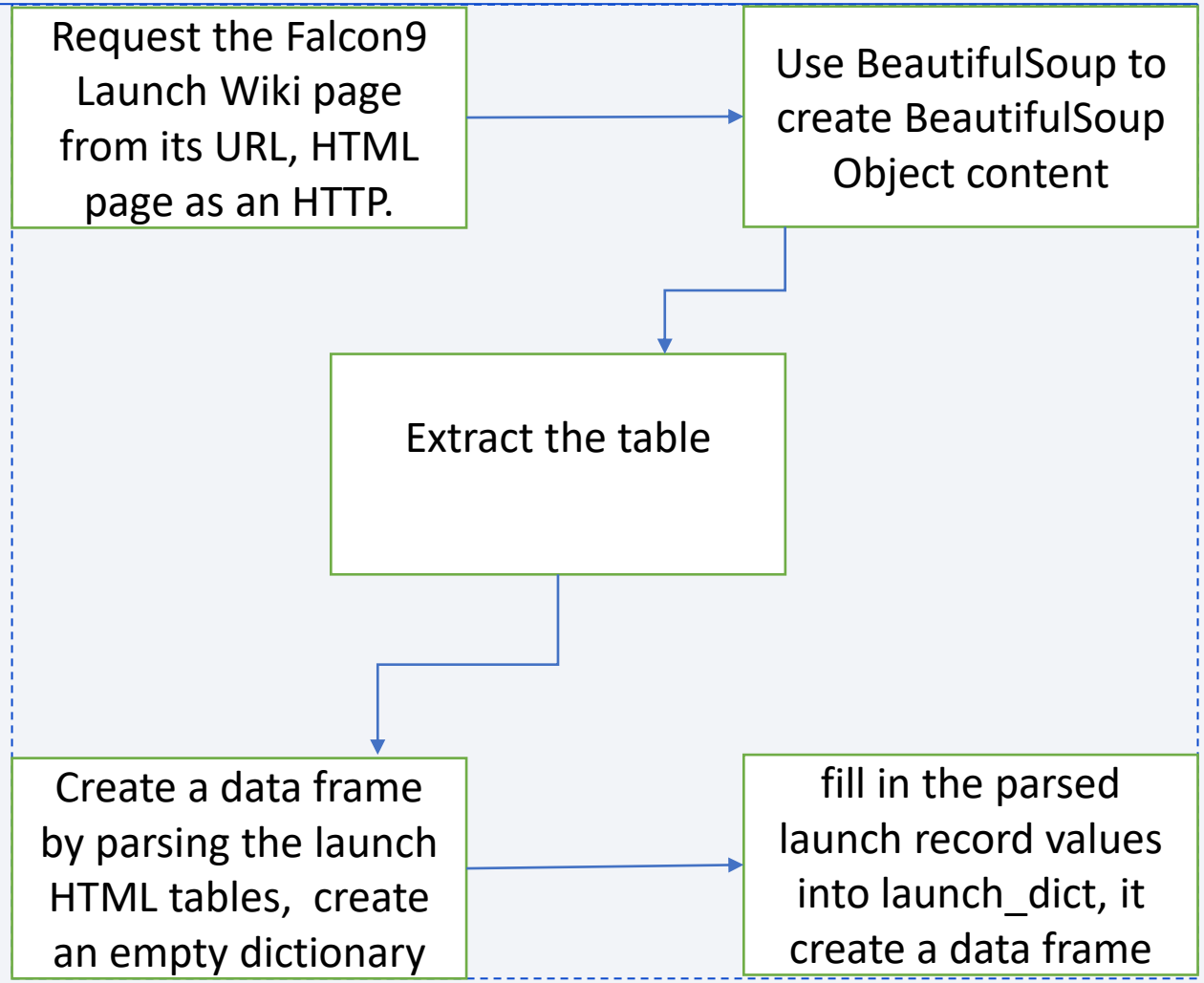


Data Collection - Scraping

- Identifying the website where data are contents. We found it on Wikipedia and extract it from there for analyzing

- Add the GitHub URL of Web Scraping:

https://github.com/williamngoy/testrepo/blob/main/webscraping2.ipynb



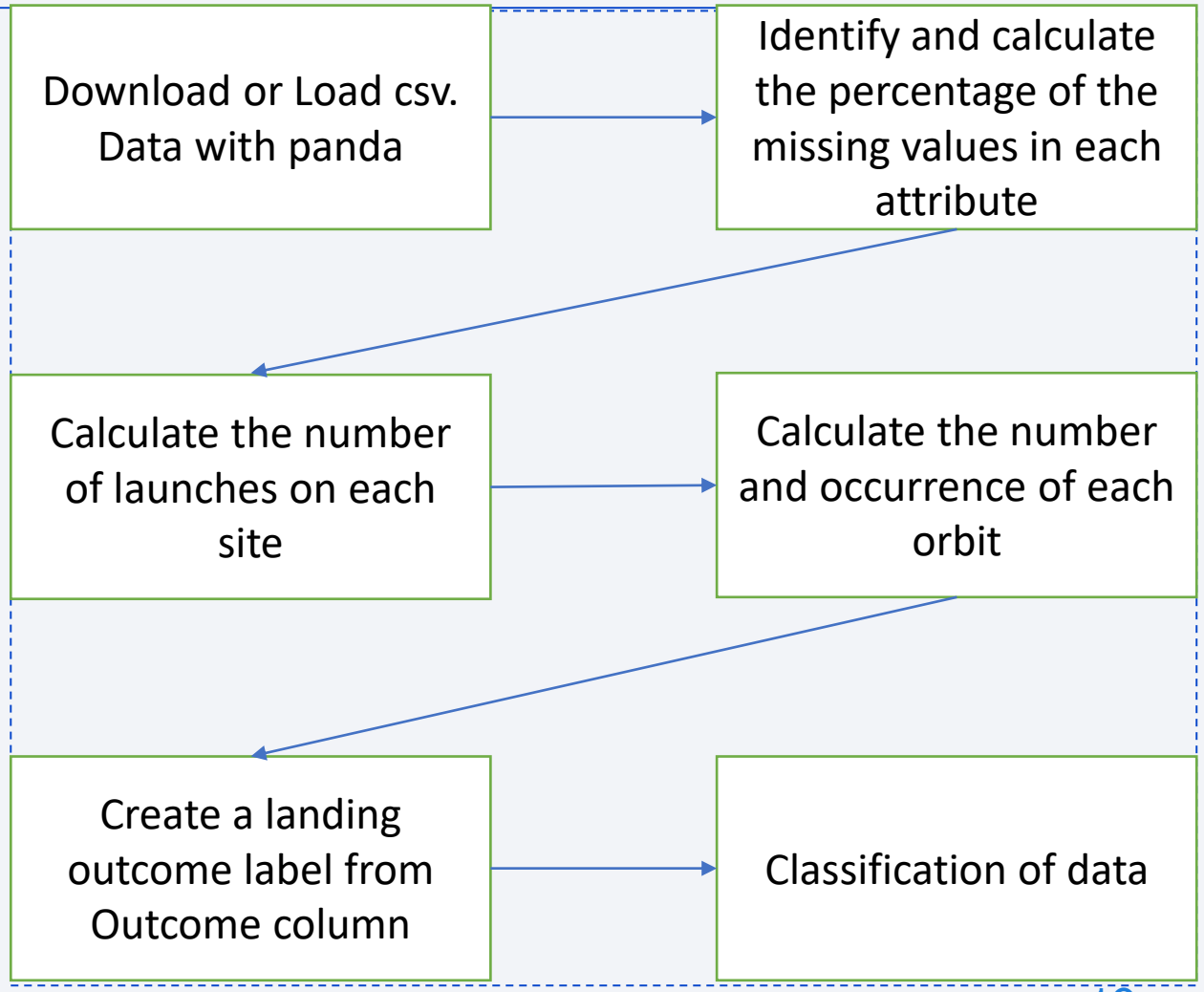
Data Wrangling

- Data wrangling is process of convert data in usable form by analyzing data.

In this process we are going to clean data and classified data for better understanding

- Add the GitHub URL of completed data wrangling:

https://github.com/williamngoy/testrepo/blob/main/data%20wrangling%2C%20data%20analysis%20(1).ipynb



EDA with Data Visualization

- Use the scatterplot to visualize the relationship between :
- Payloads Mass and Flight Number
- Launch site and Flight Number
- Payloads Mass and Launch site
- Visualize the relationship between success rate of each Orbit by using Barplot
- Visualize the relationship between Flight Number and Orbit type
- Plot line chart to see the success rate over the year
- GitHub URL of completed EDA with data visualization:

https://github.com/williamngoy/testrepo/blob/main/EDA_data_visualization.ipynb

EDA with SQL

Load the SQL extension and establish a connection with the database

- Display the names of the unique launch sites in the space mission
- Display 5 records where launch sites begin with the string 'CCA’
- Display the total payload mass carried by boosters launched by NASA (CRS)
- Display average payload mass carried by booster version F9 v1.1
- List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000
- GitHub URL of completed EDA with SQL: https://github.com/williamngoy/testrepo/blob/main/EDA_SQL.ipynb

Build an Interactive Map with Folium

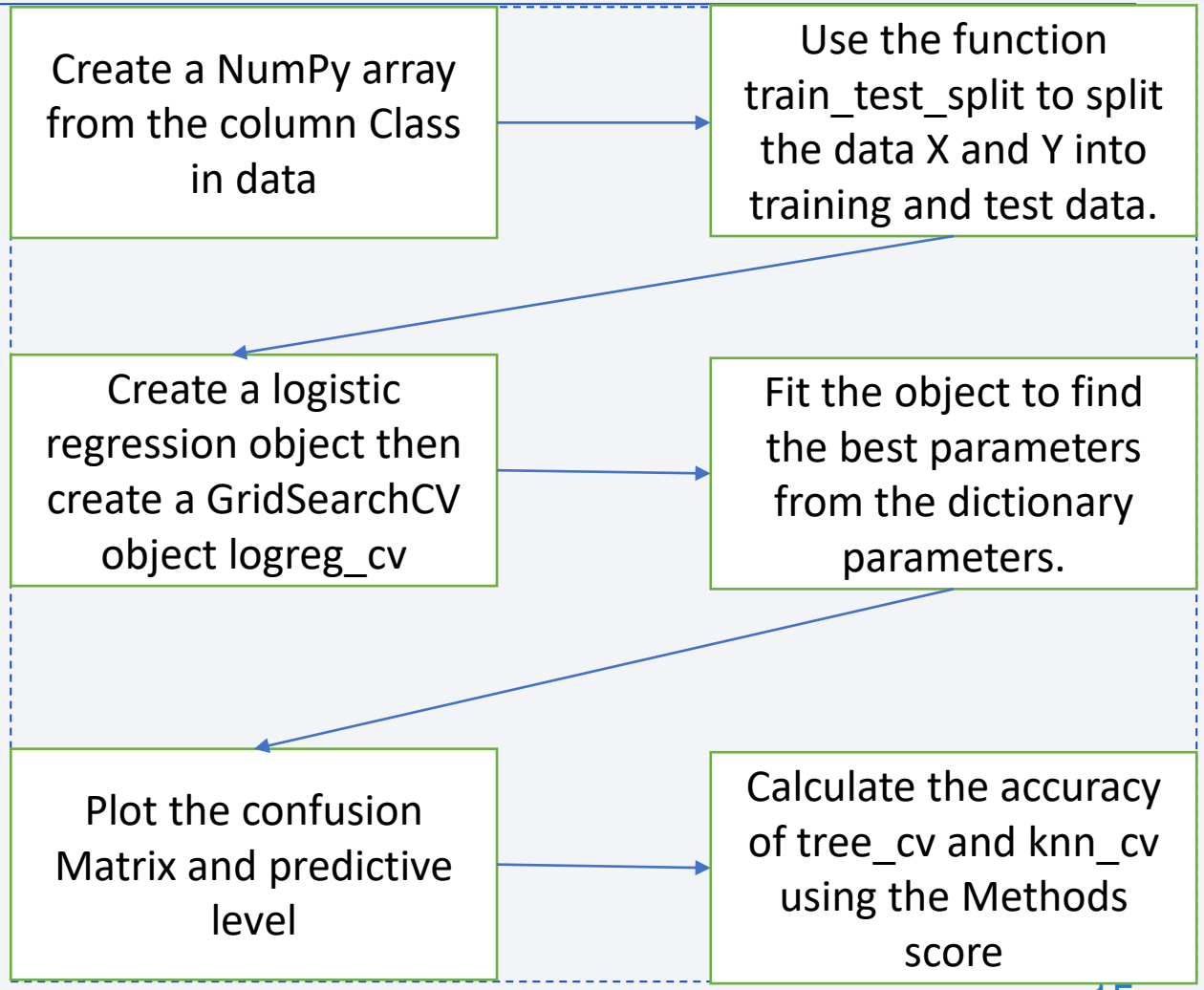
- We first need to create a folium Map object, with an initial center location to be NASA Johnson Space Center at Houston, Texas.
- Mark all launch sites on map
- Mark the success and failed of Launches for each sites on map by adding the outcome of each site
- Calculate the distances between launch sites to its proximities by exploring and analyzing the proximities of each launch site
- Find the coordinate and calculate the distance closet
- the GitHub URL of your completed interactive map with Folium map: https://github.com/williamngoy/testrepo/blob/main/launch_site_location_folium.ipynb

Build a Dashboard with Plotly Dash

- We use the Pie Chart to display :
- The successful Falcon 9 first stage distribution landings between the sites which is ;
- KSC LC-394 with 41.7 percent
- CCAFS LC-40 with 29.2 percent
- VAFB SLC-4E with 16.7 percent
- CCALC SLC-40 with 12.5 percent
- On the other hand, the scatter Plot display by class Payload Mass mission Outcome and booster version of Falcon 9
- The GitHub URL of completed Plotly Dash: https://github.com/williamngoy/ds.capstone/blob/main/Dash.html

Predictive Analysis (Classification)

- Logistic Regression, SVM (Support Vector Machine), Decision Tree, and KNN (k-Nearest Neighbors) machine learning models were trained on the training data set.
- Using the best hyper-parameters, each of the four models were scored on accuracy by using the testing data set.
- The GitHub URL of completed predictive analysis: https://github.com/williamngoy/testrepo/blob/main/Machine_Learning_Prediction.ipynb



Results

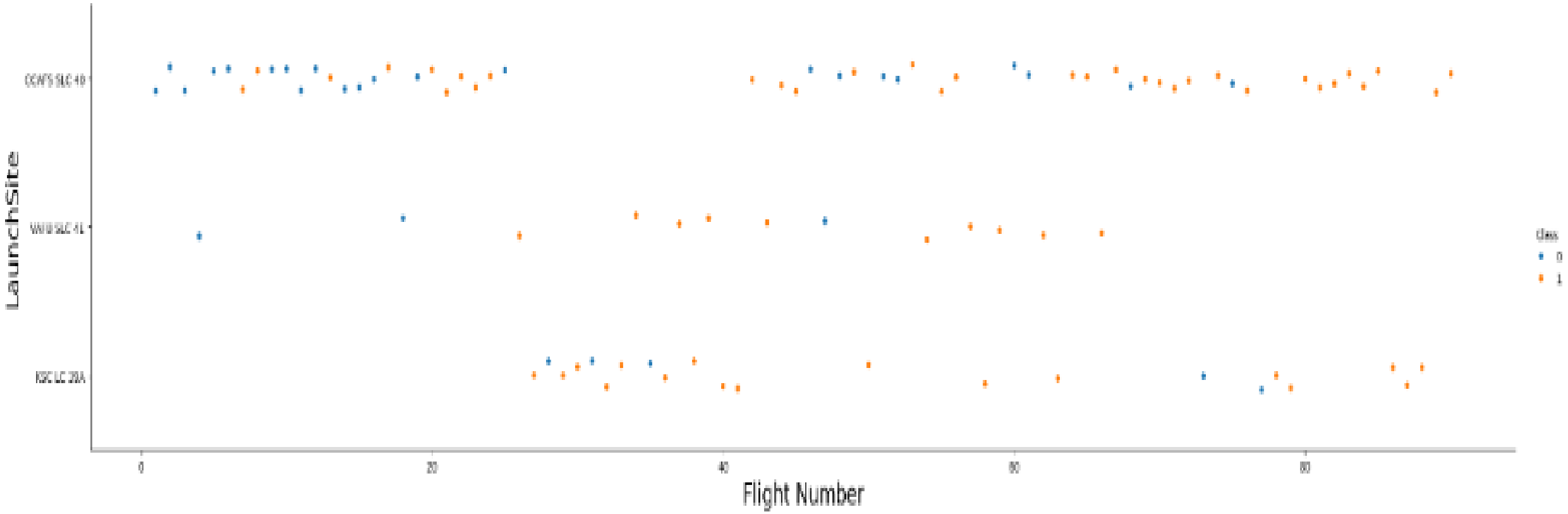
- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

Section 2

Insights drawn from EDA

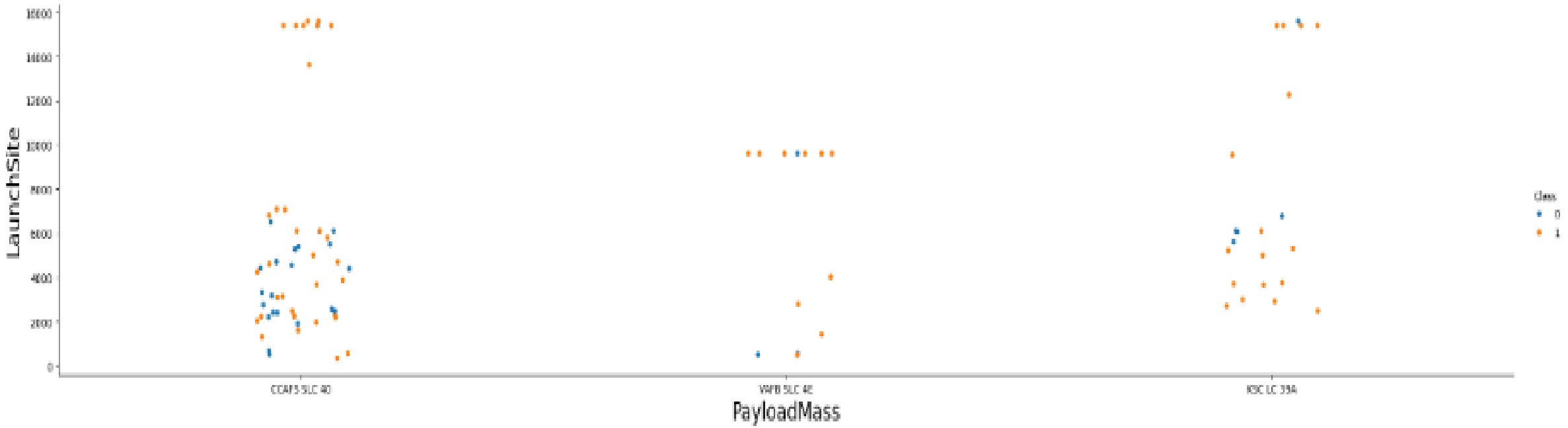
Flight Number vs. Launch Site

- The scatter plot of Flight Number vs. Launch Site shows that Falcon 9 firs stage failed landing which is indicated by (0) zero or the bleu points. And (1) successful landing in red color
- "As the flight number increases, there is a noticeable trend showing an increasing frequency of successful Falcon 9 first stage landings.



Payload vs. Launch Site

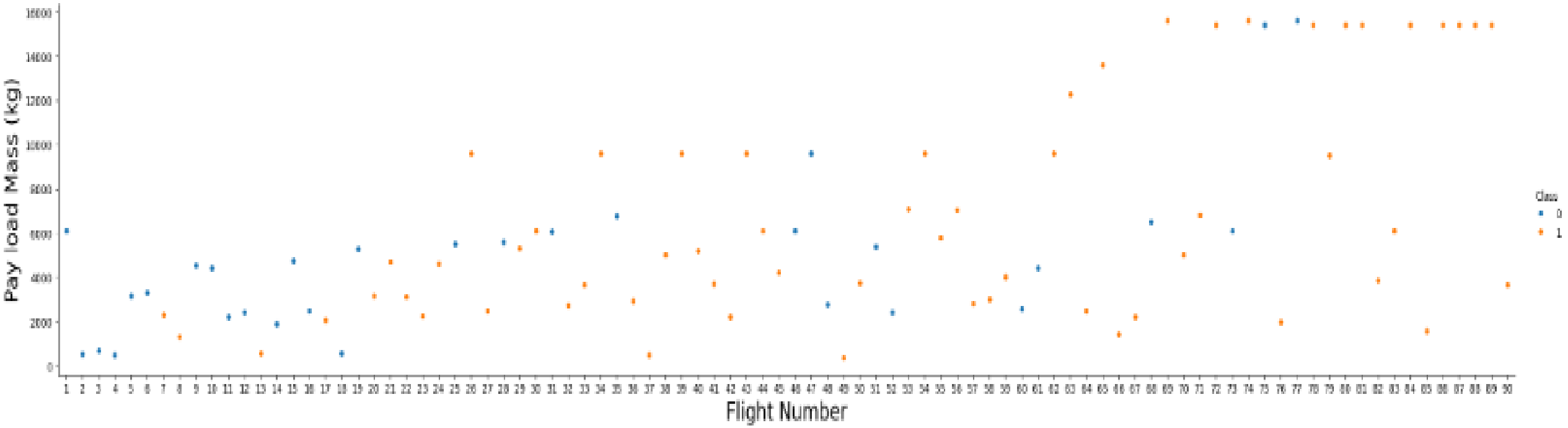
- Now if you observe Payload Vs. Launch Site scatter point chart you will find for the VAFB-SLC launch site there are no rockets launched for heavy payload mass(greater than 10000).



Now if you observe Payload Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

Payload Mass vs. Flight Number

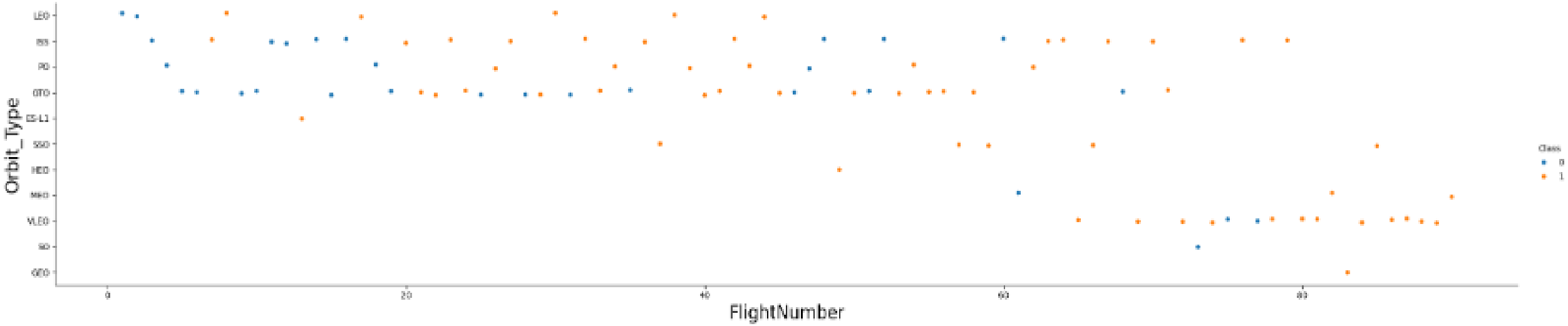
- We see that different launch sites have different success rates. CCAFS LC-40, has a success rate of 60 %, while KSC LC-39A and VAFB SLC 4E has a success rate of 77%



We see that different launch sites have different success rates. CCAFS LC-40, has a success rate of 60 %, while KSC LC-39A and VAFB SLC 4E has a success rate of 77%

Flight Number vs. Orbit Type

- For each orbit, we want to see if there is any relationship between Flight Number and Orbit type.
- You should see that in the LEO orbit the Success appears related to the number of flights; on the other hand, there seems to be no relationship between flight number when in GTO orbit.



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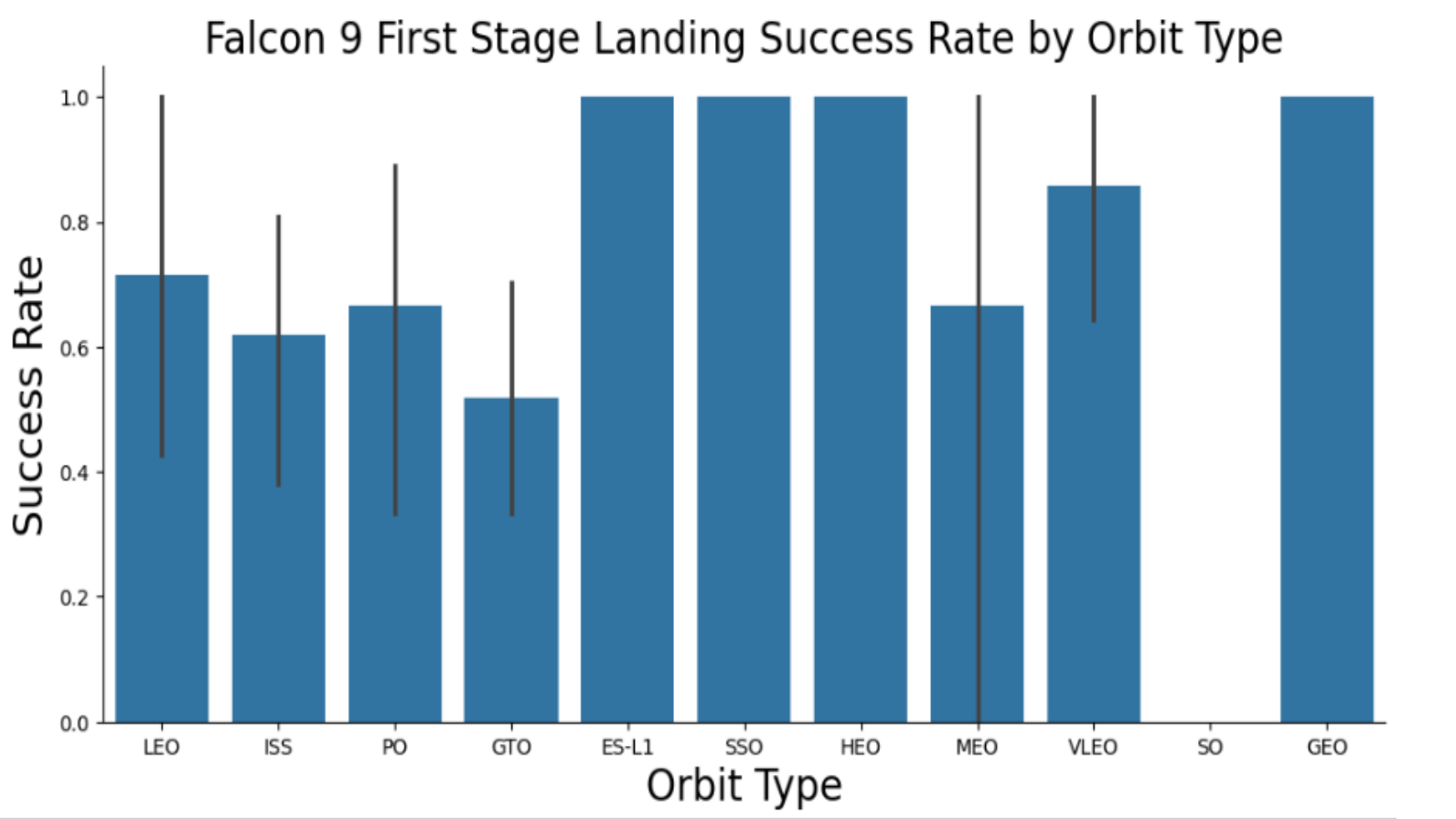
You should see that in the LEO orbit the Success appears related to the number of flights; on the other hand, there seems to be no relationship between flight number when in GTO orbit.

Success Rate vs. Orbit Type

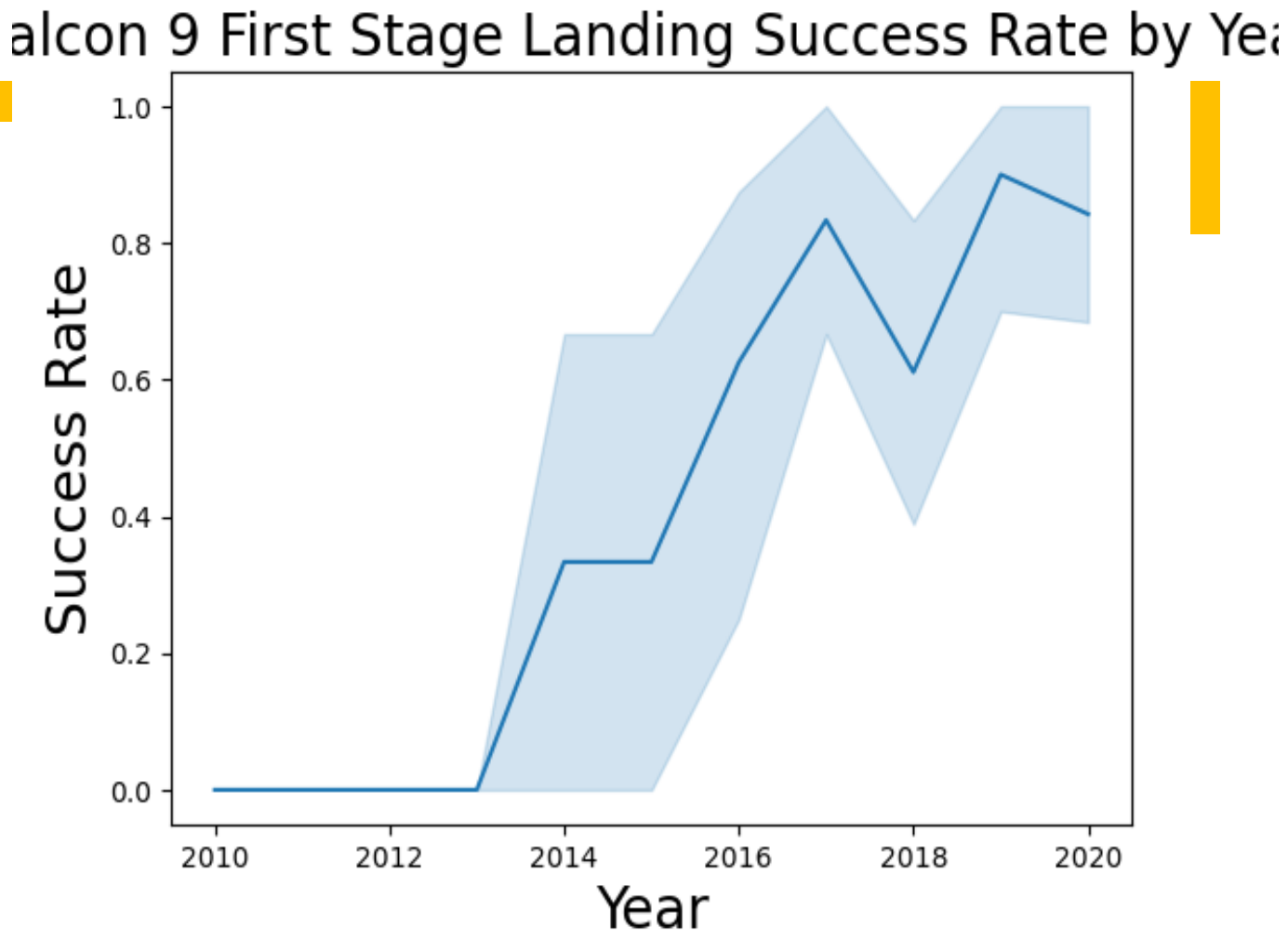
- We can notice that ES-L1, SSO, HEO and GEO orbits have no failed first stage landings.
- •and SO orbits have no successful first stage landings.



Launch Success Yearly Trend

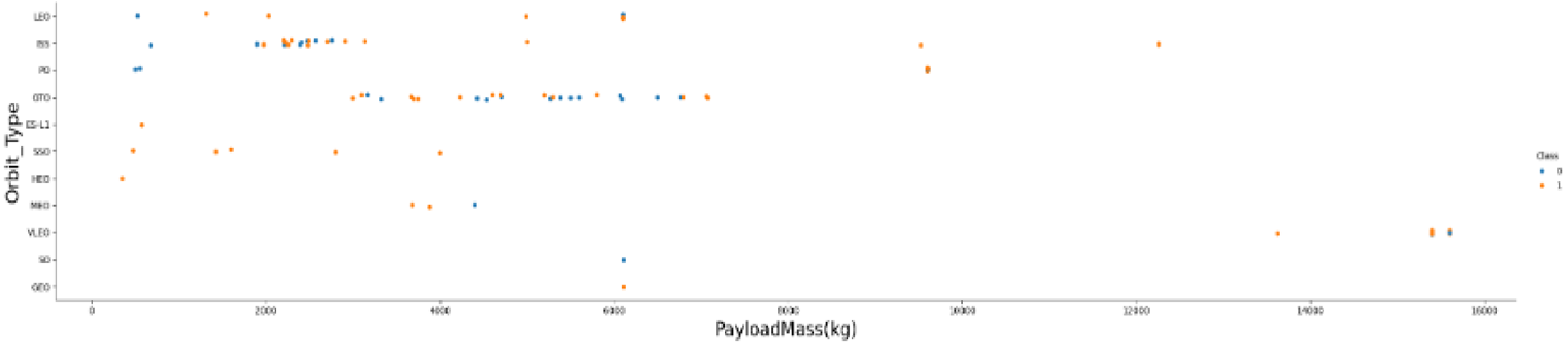
- The success rate has increased significantly over the years.

- You can observe that the success rate since 2013 kept increasing till 2020



Payload vs. Orbit Type

- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.
- However, for GTO we cannot distinguish this well as both positive landing rate and negative landing(unsuccessful mission) are both there here



Similarly, we can plot the Payload vs. Orbit scatter point charts to reveal the relationship between Payload and Orbit type

With heavy payloads the successful landing or positive landing rate are more for Polar,LEO and ISS.

However for GTO we cannot distinguish this well as both positive landing rate and negative landing(unsuccessful mission) are both there here

All Launch Site Names

```
%sql SELECT DISTINCT LAUNCH_SITE FROM SPACEXTBL;
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

There is four launch sites as shown on the picture

Launch Site Names Begin with 'CCA'

- Here are 5 records where launch sites begin with `CCA`

```
]: %sql SELECT * FROM SPACEXTBL WHERE (LAUNCH_SITE) LIKE 'CCA%' LIMIT 5;
 * sqlite:///my_data1.db
Done.
```

]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
6/4/2010	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
12/8/2010	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
22/05/2012	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
10/8/2012	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
3/1/2013	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- We calculate the total payload carried by boosters from NASA

```
%sql select sum(PAYLOAD_MASS__KG_) as total_payload_mass from SPACEXTBL where customer = 'NASA (CRS)';
```

```
 * sqlite:///my_data1.db

Done.
```

total_payload_mass
45596

Average Payload Mass by F9 v1.1

- The average payload mass carried by booster version F9 v1.1

```
%sql select avg(PAYLOAD_MASS__KG_) as Average_payload_mass from SPACEXTBL where Booster_Version = 'F9 v1.1';
```

```
 * sqlite:///my_data1.db
Done.
```

Average_payload_mass
2928.4

First Successful Ground Landing Date

- Here is the first date of the first successful landing outcome on ground pad

```
# This is the full record of the first successful landing outcome in ground pad
%sql SELECT * FROM SPACEXTBL WHERE DATE = '1/10/2015';
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
1/10/2015	9:47:00	F9 v1.1 B1012	CCAFS LC-40	SpaceX CRS-5	2395	LEO (ISS)	NASA (CRS)	Success	Failure (drone ship)

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
%%sql
SELECT DISTINCT(BOOSTER_VERSION), LANDING__OUTCOME, PAYLOAD_MASS__KG_
FROM SPACEXTBL
WHERE LANDING__OUTCOME = 'Success (drone ship)' AND PAYLOAD_MASS__KG_ BETWEEN 4000 AND 6000;
```

```
 * ibm_db_sa://wvx02184:***@2f3279a5-73d1-4859-88f0-a6c3e6b4b907.c3n41cmd0nqnrk39u98g.databases.appdomain.cloud:3075
6/bludb
Done.
```

booster_version	landing__outcome	payload_mass__kg_
F9 FT B1021.2	Success (drone ship)	5300
F9 FT B1031.2	Success (drone ship)	5200
F9 FT B1022	Success (drone ship)	4696
F9 FT B1026	Success (drone ship)	4600

Total Number of Successful and Failure Mission Outcomes

- The total number of successful and failure mission outcomes

```
%sql select count(*) as Mission_Outcome from SPACEXTBL where Landing_Outcome NOT LIKE '%success%' OR Landing_Outcome LIKE '%success%';
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome
101

```
%sql select count(*) as Mission_Outcome from SPACEXTBL where Landing_Outcome LIKE '%success%'
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome
61

```
%sql select count(*) as Mission_Outcome from SPACEXTBL where Landing_Outcome NOT LIKE '%success%'
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome
40

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass

```
%%sql
SELECT DISTINCT(BOOSTER_VERSION), PAYLOAD_MASS__KG_
FROM SPACEXTBL
WHERE PAYLOAD_MASS__KG_ = (SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL)
```

```
 * ibm_db_sa://wvx02184:***@2f3279a5-73d1-4859-88f0-a6c3e6b4b907.c3n41cmd0nqnrk39u98g.databases.appdomain.cloud:3075
6/bludb
Done.
```

booster_version	payload_mass__kg_
F9 B5 B1048.4	15600
F9 B5 B1048.5	15600
F9 B5 B1049.4	15600
F9 B5 B1049.5	15600
F9 B5 B1049.7	15600
F9 B5 B1051.3	15600
F9 B5 B1051.4	15600
F9 B5 B1051.6	15600
F9 B5 B1056.4	15600
F9 B5 B1058.3	15600
F9 B5 B1060.2	15600
F9 B5 B1060.3	15600

2015 Launch Records

- List the failed landing outcomes in drone ship, their booster versions, and launch site names for in year 2015

List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

```
%%sql
SELECT LANDING__OUTCOME, BOOSTER_VERSION, LAUNCH_SITE, YEAR(DATE) AS DATE_YEAR
FROM SPACEXTBL
WHERE LANDING__OUTCOME = 'Failure (drone ship)' AND YEAR(DATE) = '2015'
```

```
 * ibm_db_sa://wvx02184:***@2f3279a5-73d1-4859-88f0-a6c3e6b4b907.c3n41cmd0nqnrk39u98g.databases.appdomain.cloud:3075
6/bludb
Done.
```

landing__outcome	booster_version	launch_site	date_year
Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40	2015
Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40	2015

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

```
%%sql
SELECT LANDING__OUTCOME, COUNT(LANDING__OUTCOME) AS COUNT
FROM SPACEXTBL
WHERE DATE BETWEEN '2010-06-04'  AND '2017-03-20'
GROUP BY LANDING__OUTCOME
ORDER BY COUNT DESC
```

```
 * ibm_db_sa://wvx02184:***@2f3279a5-73d1-4859-88f0-a6c3e6b4b907.c3n41cmd0nqnrk39u98g.databases.appdomain.cloud:3075
6/bludb
Done.
```

landing__outcome	COUNT
No attempt	10
Failure (drone ship)	5
Success (drone ship)	5
Controlled (ocean)	3
Success (ground pad)	3
Failure (parachute)	2
Uncontrolled (ocean)	2
Precluded (drone ship)	1

Section 3

Launch Sites Proximities Analysis

Launch site location Falcon9

- On this screem we can see the map of all launch site in the usa

•VAFB SLC-4E (California, USA)

• Vandenberg Air Force Base Space Launch Complex 4E

•KSC LC-39A (Florida, USA)

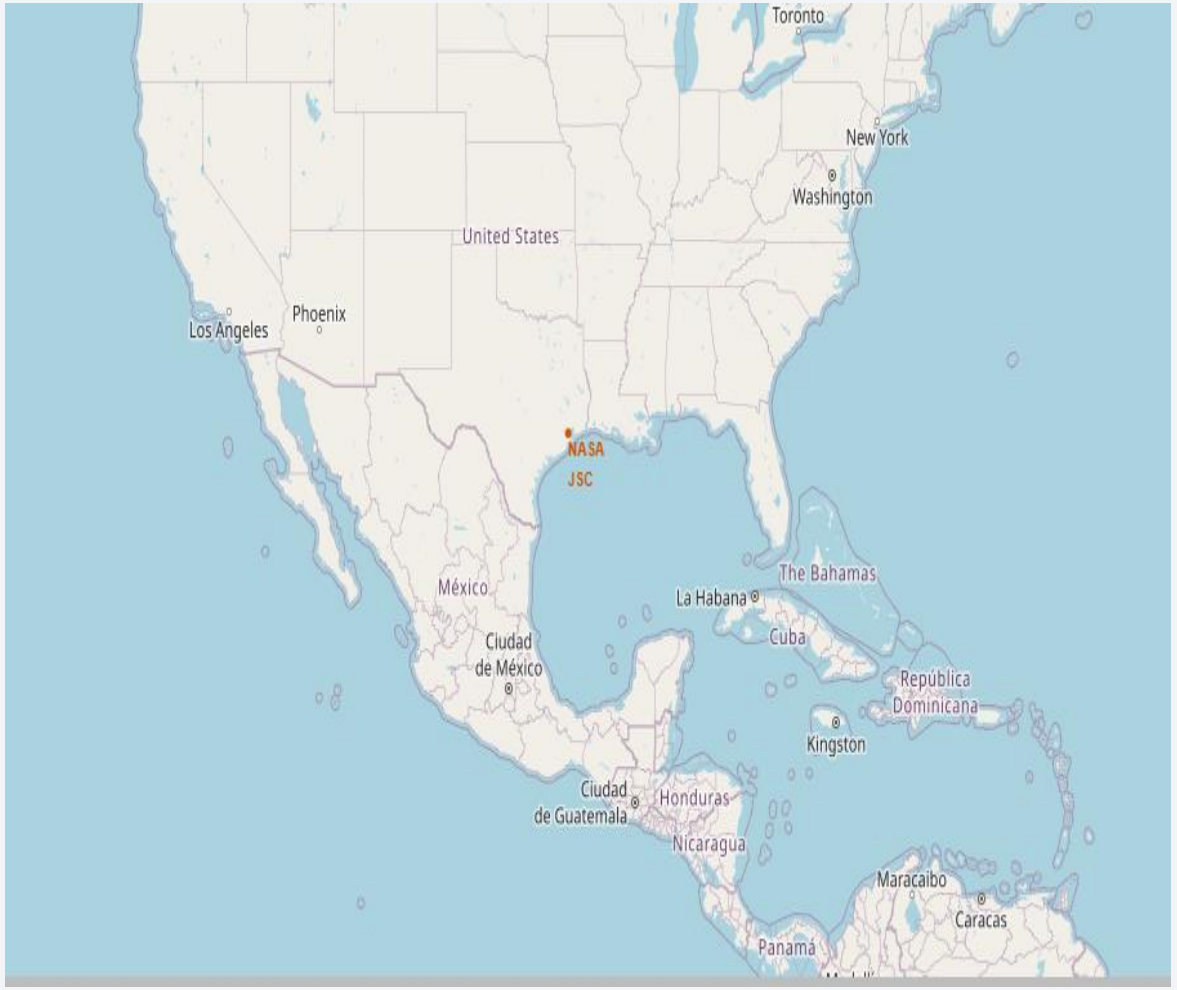
• Kennedy Space Center Launch Complex 39A

•CCAFS LC-40 (Florida, USA)

• Cape Canaveral Air Force Station Launch Complex 40

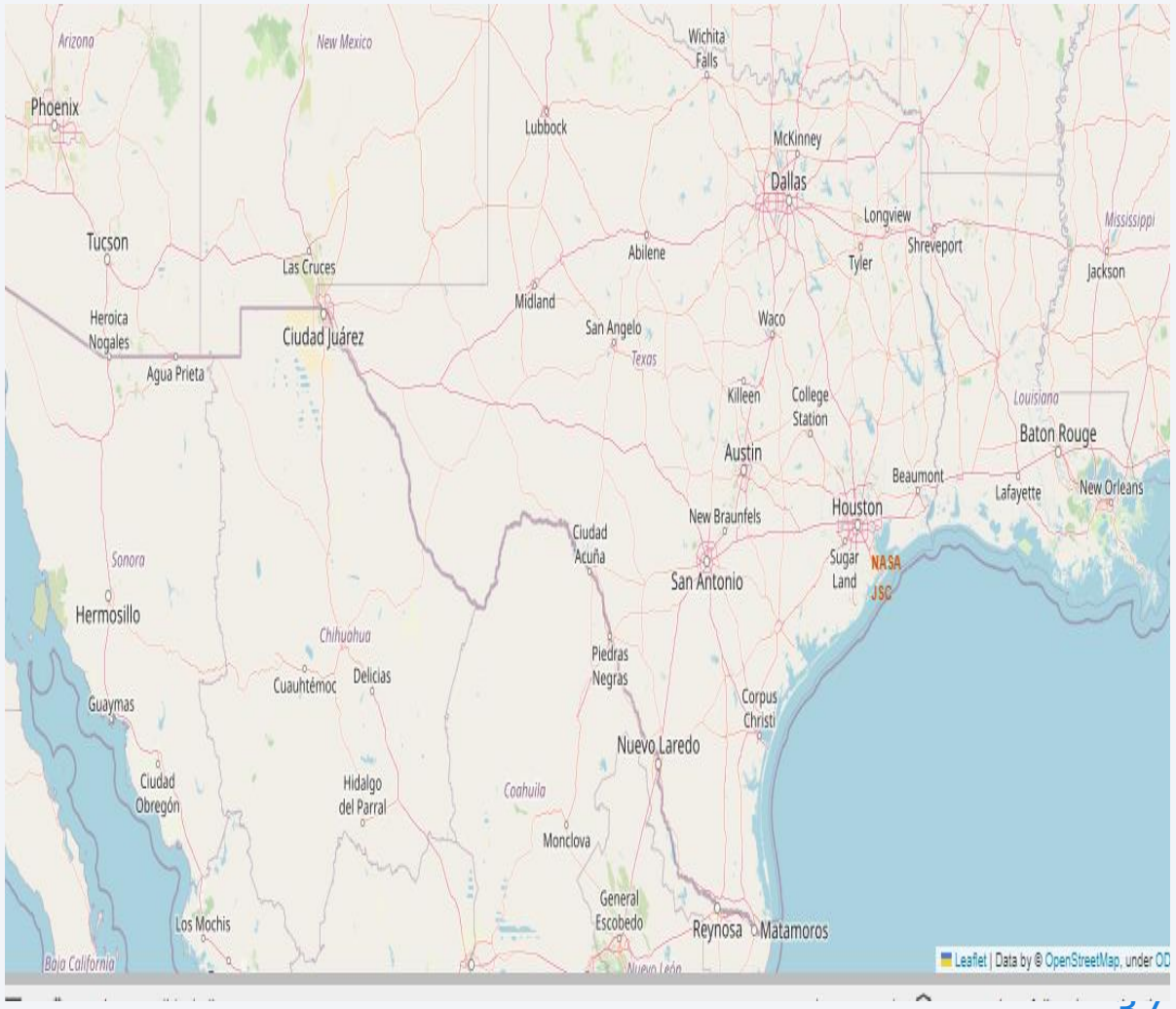
•CCAFS SLC-40 (Florida, USA)

• Cape Canaveral Air Force Station Space Launch Complex 40



Marker coordinate of each launch site

	Launch Site	Lat	Long	class
46	KSC LC-39A	28.573255	-80.646895	1
47	KSC LC-39A	28.573255	-80.646895	1
48	KSC LC-39A	28.573255	-80.646895	1
49	CCAFS SLC-40	28.563197	-80.576820	1
50	CCAFS SLC-40	28.563197	-80.576820	1
51	CCAFS SLC-40	28.563197	-80.576820	0
52	CCAFS SLC-40	28.563197	-80.576820	0
53	CCAFS SLC-40	28.563197	-80.576820	0
54	CCAFS SLC-40	28.563197	-80.576820	1
55	CCAFS SLC-40	28.563197	-80.576820	0



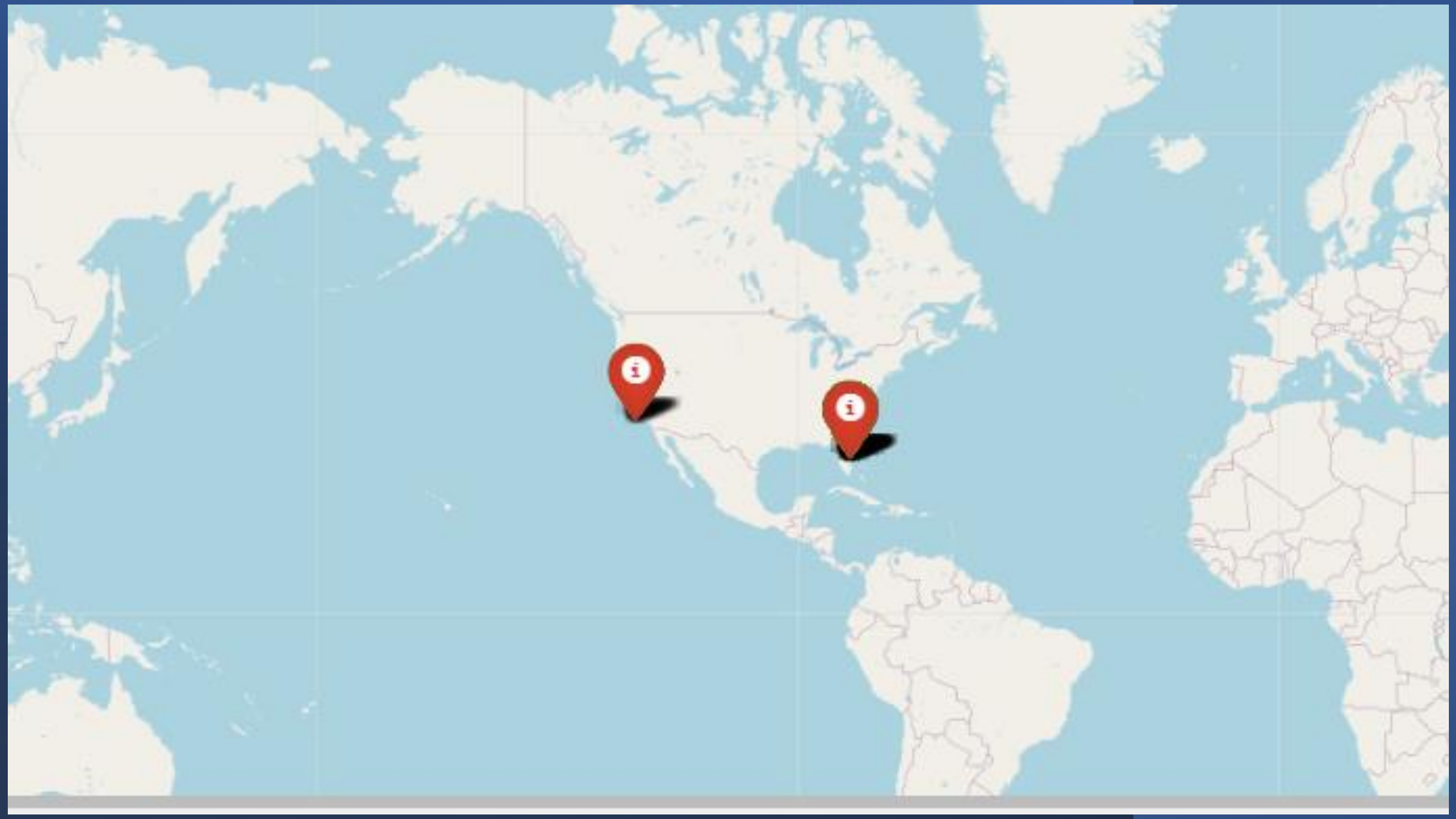
Distance coordinate for each launch site

- Distance

```
[25]: # Coordinates of the closet RAIL LINE
rail_line_lat = 28.57209
rail_line_lon = -80.58520
distance_rail_line = calculate_distance(launch_site_lat, launch_site_lon, rail_line_lat, rail_line_lon)
print(f'Distance to Rail Line: {round(distance_rail_line,2)} km')

# Create and add a folium.Marker on the selected point on the map
distance_marker = folium.Marker(
    [rail_line_lat, rail_line_lon],
    icon=DivIcon(
        icon_size=(20,20),
        icon_anchor=(-40,-20),
        html=f'<div style="font-size:12px; color:#222; font-weight:bold; width:200px;">Rail Line {round(dist
        )
    )

Distance to Rail Line: 1.33 km
```

```
[24]: # Coordinates of the closet COASTLINE
coastline_lat = 28.56321
coastline_lon = -80.56801
distance_coastline = calculate_distance(launch_site_lat, launch_site_lon, coastline_lat, coastline_lon)
print(f'Distance to Coastline: {round(distance_coastline,2)} km')

# Create and add a folium.Marker on the selected point on the map
distance_marker = folium.Marker(
    [coastline_lat, coastline_lon],
    icon=DivIcon(
        icon_size=(20,20),
        icon_anchor=(100,-10),
        html=f'<div style="font-size:12px; color:#222; font-weight:bold; width:200px;">Coastline {round(distance_coastline,2)} km</d
        )
    )

Distance to Coastline: 0.92 km
```

The CCAFS LC-40 and CCAFS SLC-40 launch sites have coordinates that are close to being, but are not exactly, right on top of each other. •The perimeter road around CCAFS LC-40 is 0.19 km away from the launch site coordinates. •The coastline is 0.92 km away from CCAFS LC-40. •The rail line is 1.33 km away from CCAFS LC-40.

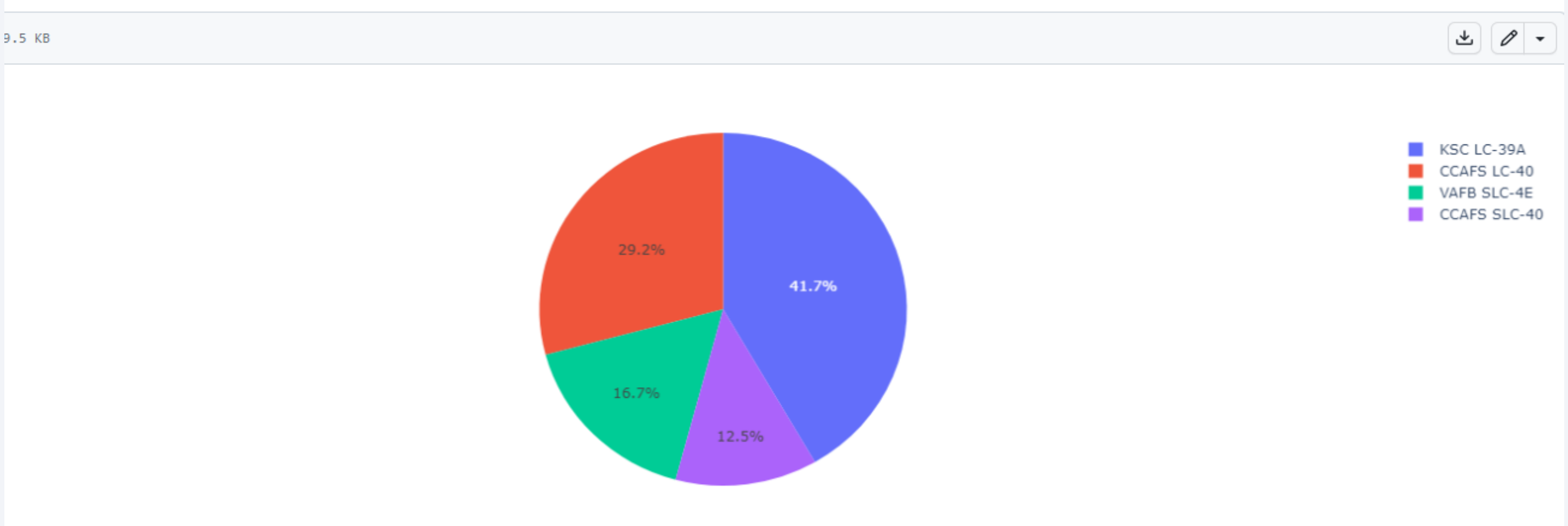


Section 4

Build a Dashboard with Plotly Dash

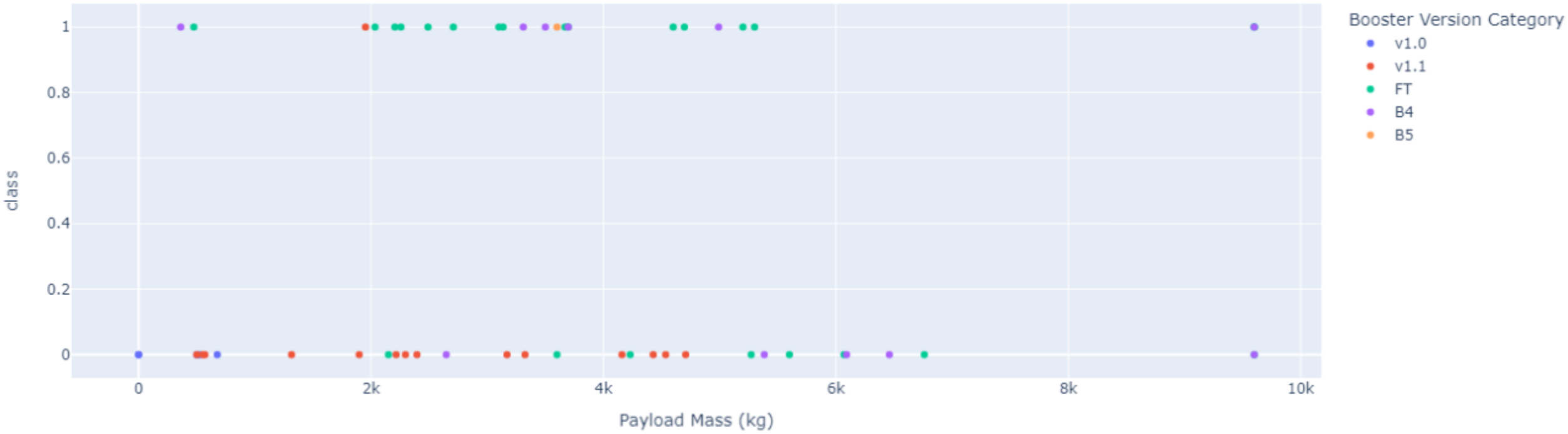
SpaceX launch record dashboard

- The greatest share of successful
- Falcon 9 first stage landing
- outcomes (at 41.7% of the total)
- occurred at KSC LC-39A.



Payload and launch outcome

- The payload range from
- about 2,000 kg to 5,000 kg
- has the largest success
- rate.



Section 5

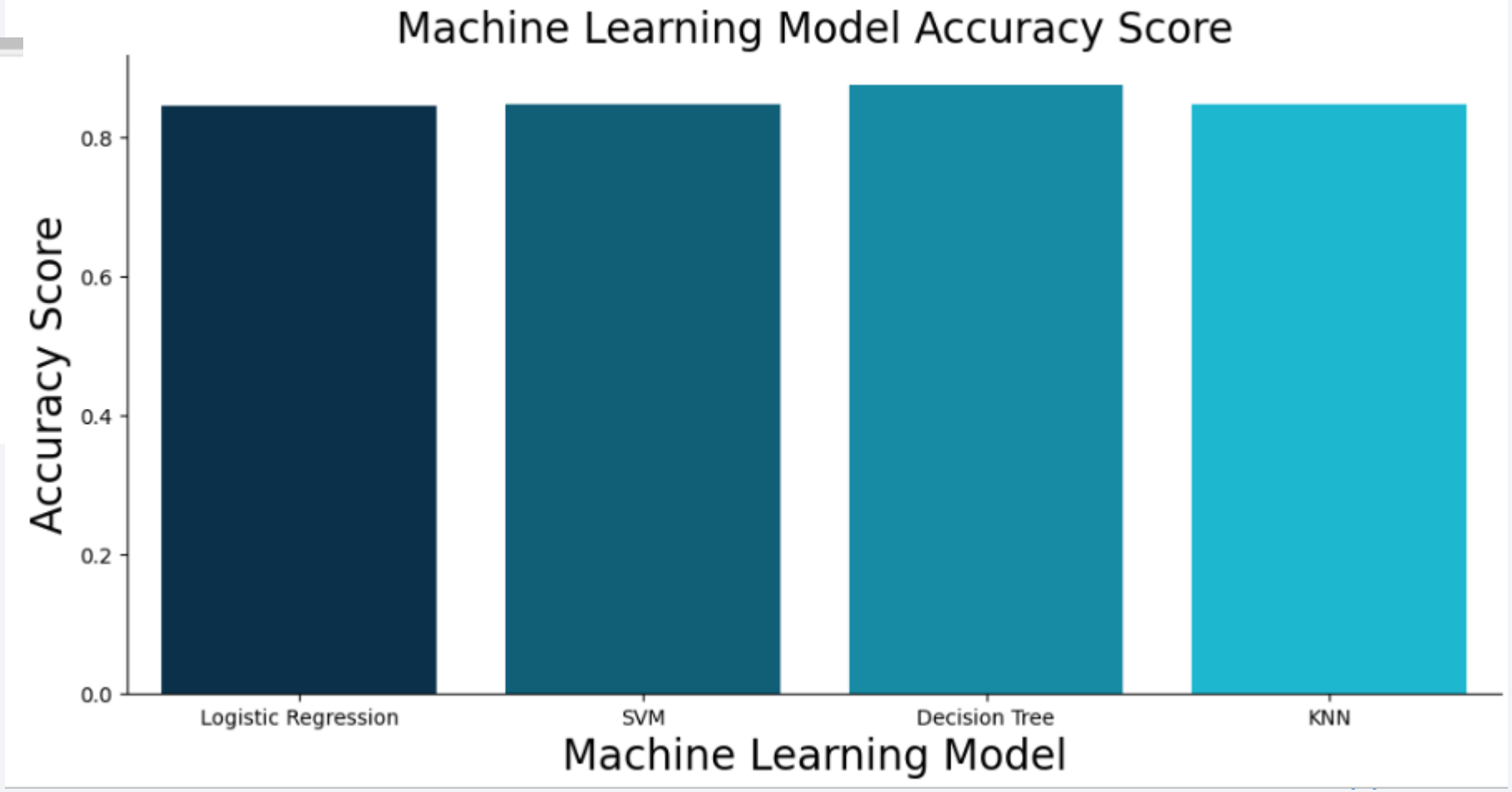
Predictive Analysis (Classification)

Classification Accuracy

- Here is the accuracy of the model and the performance

1]:

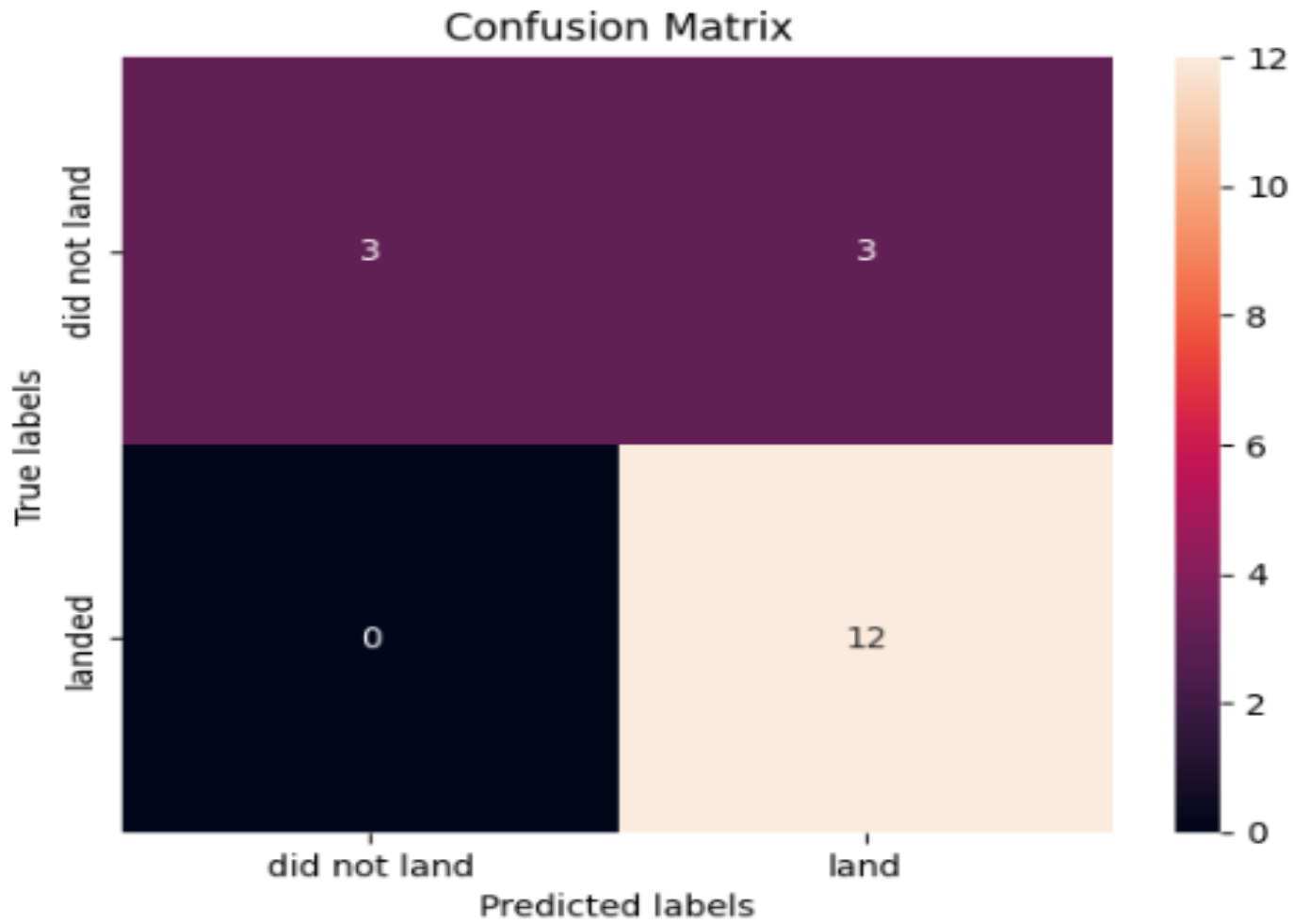
	Model	Score	Max Score
0	Logistic Regression	0.846429	0
1	SVM	0.848214	0
2	Decision Tree	0.875000	Max Score
3	KNN	0.848214	0



Confusion Matrix

- Here is the confusion matrix of logistic regression
- On this matrix we can see that the prediction says that 12 True positive and 3False positive. We also have 3false positive and 0 false Negative

```
yhat=logreg_cv.predict(X_test)
logreg_cv.score(X_test, Y_test)
plot_confusion_matrix(Y_test,yhat)
```



Conclusions

- Machine landing model can be use for prediction about next landing for falcon 9
- SpaceX falcon 9 as the most success landing stage than other falcon

Appendix

- **Jupyter notebook/jupyter lab files**
- https://github.com/williamngoy/testrepo/blob/main/data-collection-api%20(1).ipynb
- https://github.com/williamngoy/testrepo/blob/main/data%20wrangling%2C%20data%20analysis%20(1).ipynb
- https://github.com/williamngoy/testrepo/blob/main/EDA_data_visualization.ipynb
- https://github.com/williamngoy/testrepo/blob/main/EDA_SQL.ipynb
- https://github.com/williamngoy/testrepo/blob/main/launch_site_location_folium.ipynb
- https://github.com/williamngoy/testrepo/blob/main/Machine_Learning_Prediction.ipynb
- https://github.com/williamngoy/testrepo/blob/main/webscraping2.ipynb
- https://github.com/williamngoy/testrepo/blob/main/Dash.html

Png files

- https://github.com/williamngoy/testrepo/blob/main/Machine_Learning_model_accuracy.PNG
- https://github.com/williamngoy/testrepo/blob/main/Success_rate%20vs%20Year.PNG
- https://github.com/williamngoy/testrepo/blob/main/newplot%20(1).png
- https://github.com/williamngoy/testrepo/blob/main/newplot.png

Thank you!